

ECE/CS 757 Advanced Computer Architecture II

Spring 2025

Multicores and Multiprogramming

Instructor: Prof. Joshua San Miguel

Some slides adopted from Greg Steffan and Natalie Enright Jerger

Types of Parallelism

Compute-Level Parallelism:

- Executing multiple computation streams simultaneously

Data-Level Parallelism:

- Processing multiple data streams simultaneously

*-Level Parallelism

Request-Level Parallelism:

- Users issue simultaneous requests (e.g., databases, web servers, transactional systems)
- Compute-level parallelism in multi-chip processors

Task-Level Parallelism:

- Programs invoke simultaneous tasks (e.g., dataflow, message passing, speculative processors)
- Compute-level parallelism in multi-chip processors and multiprocessors

Thread-Level Parallelism:

- Programs execute simultaneous threads (e.g., pthreads, OpenMP, GPUs)
- Compute-level parallelism in multiprocessors

Memory-Level Parallelism:

- Threads issue simultaneous memory accesses (e.g., cache misses, prefetchers)
- Data-level parallelism in multiprocessors

*-Level Parallelism

Instruction-Level Parallelism:

- Threads issue simultaneous instructions (e.g., out-of-order, superscalar)
- Compute-level parallelism in uniprocessors

Superword-Level Parallelism:

- Instructions process simultaneous data words (e.g., vector instructions, Intel SSE/AVX)
- Data-level parallelism in uniprocessors

Bit-Level Parallelism:

- Operations manipulate simultaneous bits (e.g., functional units, bitwise arithmetic)
- Data-level parallelism in uniprocessors

Parallelism and Concurrency

Concurrency:

Coexistence of multiple tasks/computations in progress

- Logically doing multiple things at once
- Typically algorithm-defined

Parallelism:

Execution of multiple tasks/computations simultaneously

- Physically doing multiple things at once
- Typically architecture-defined

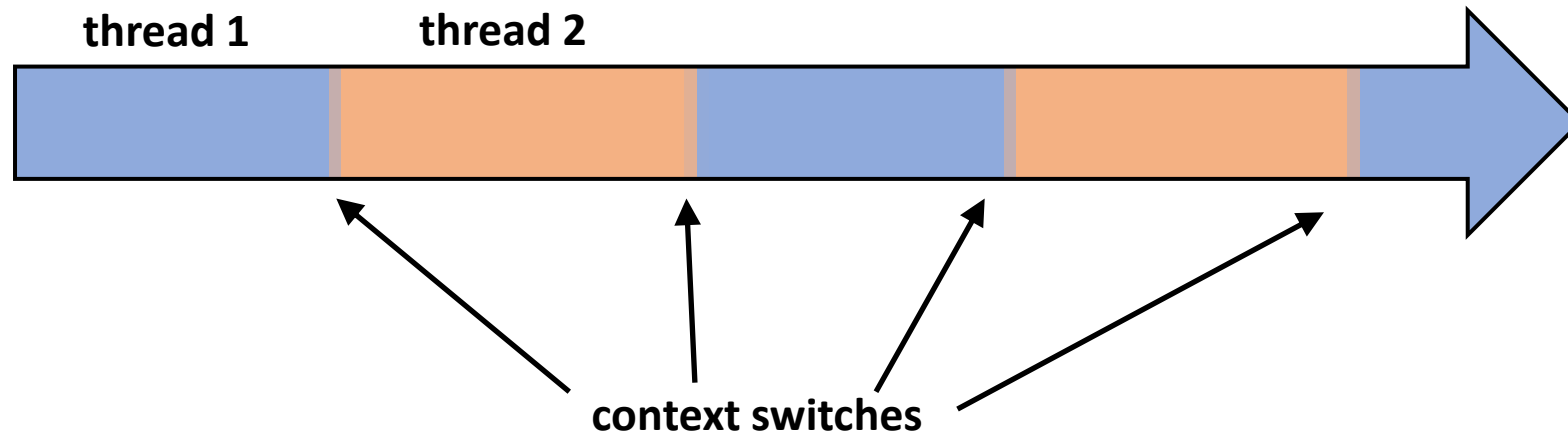
Note that different views/definitions exist depending on who you talk to.



Parallelism and Concurrency

e.g., Concurrent but not parallel:

- Context-switching threads on a single core



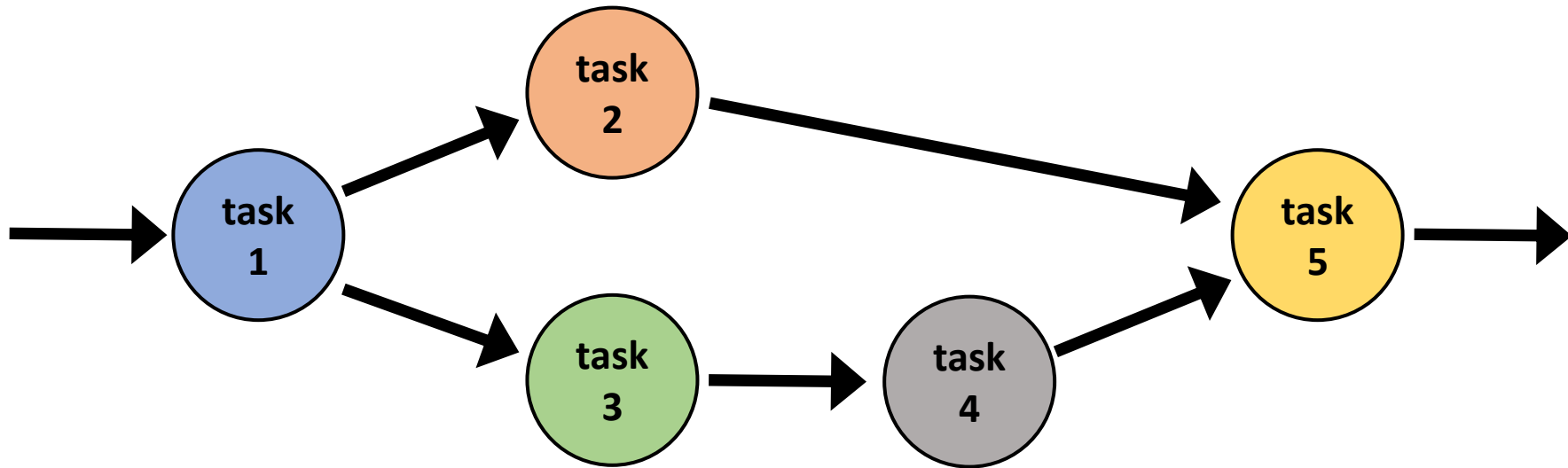
Parallelism and Concurrency

e.g., Parallel but not concurrent:

Parallelism and Concurrency

e.g., Parallel but not concurrent:

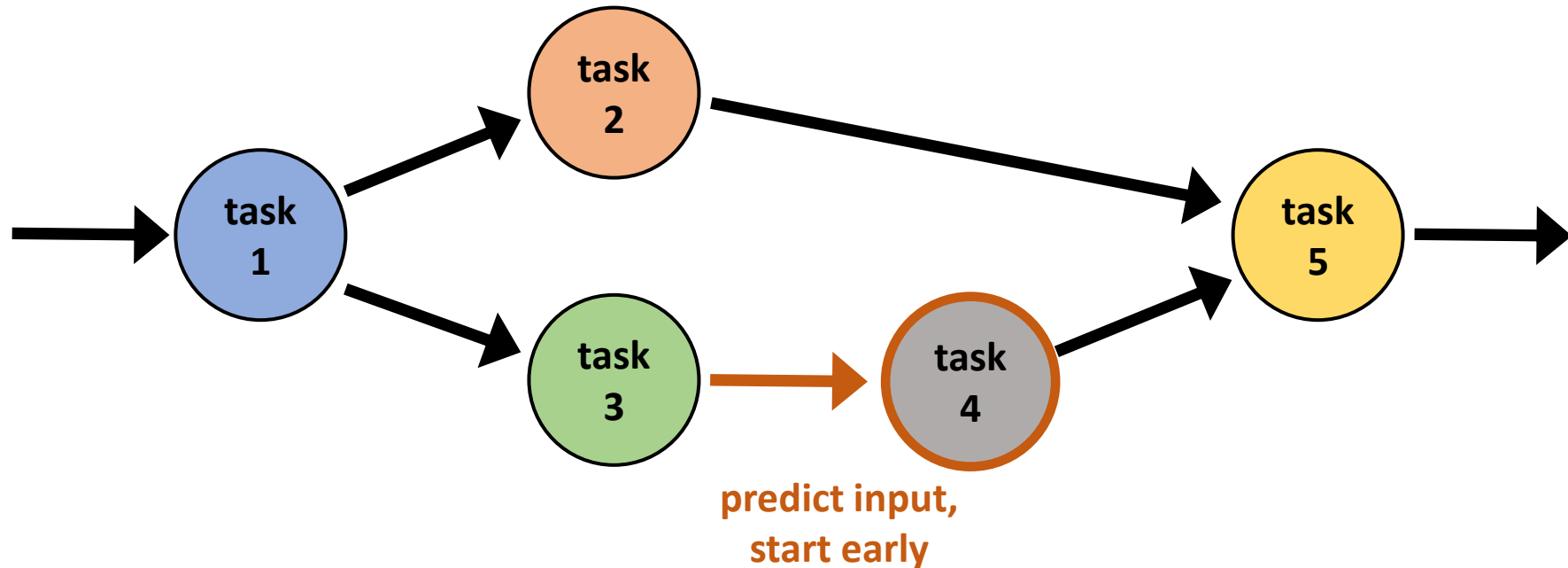
- Speculative task parallelism



Parallelism and Concurrency

e.g., Parallel but not concurrent:

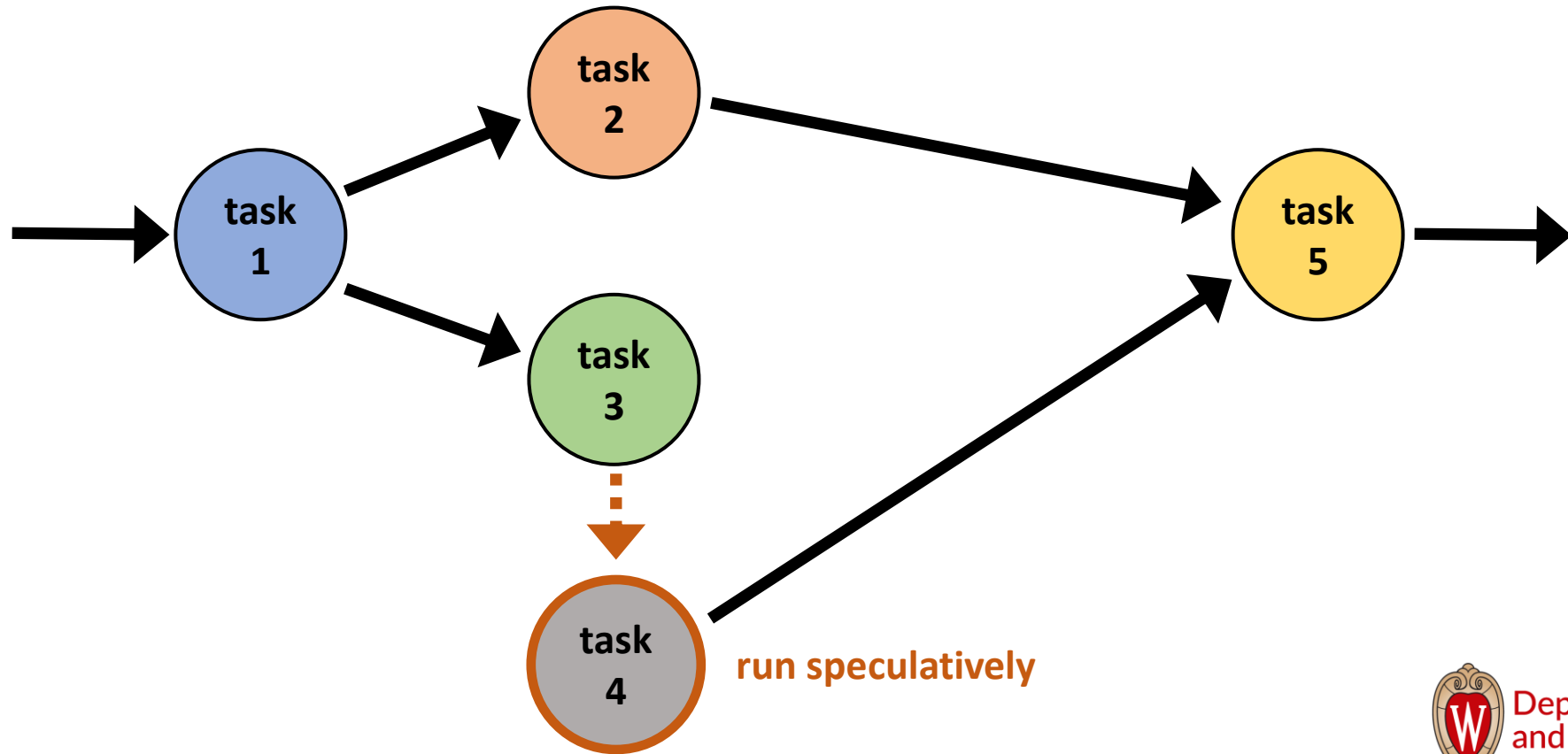
- Speculative task parallelism



Parallelism and Concurrency

e.g., Parallel but not concurrent:

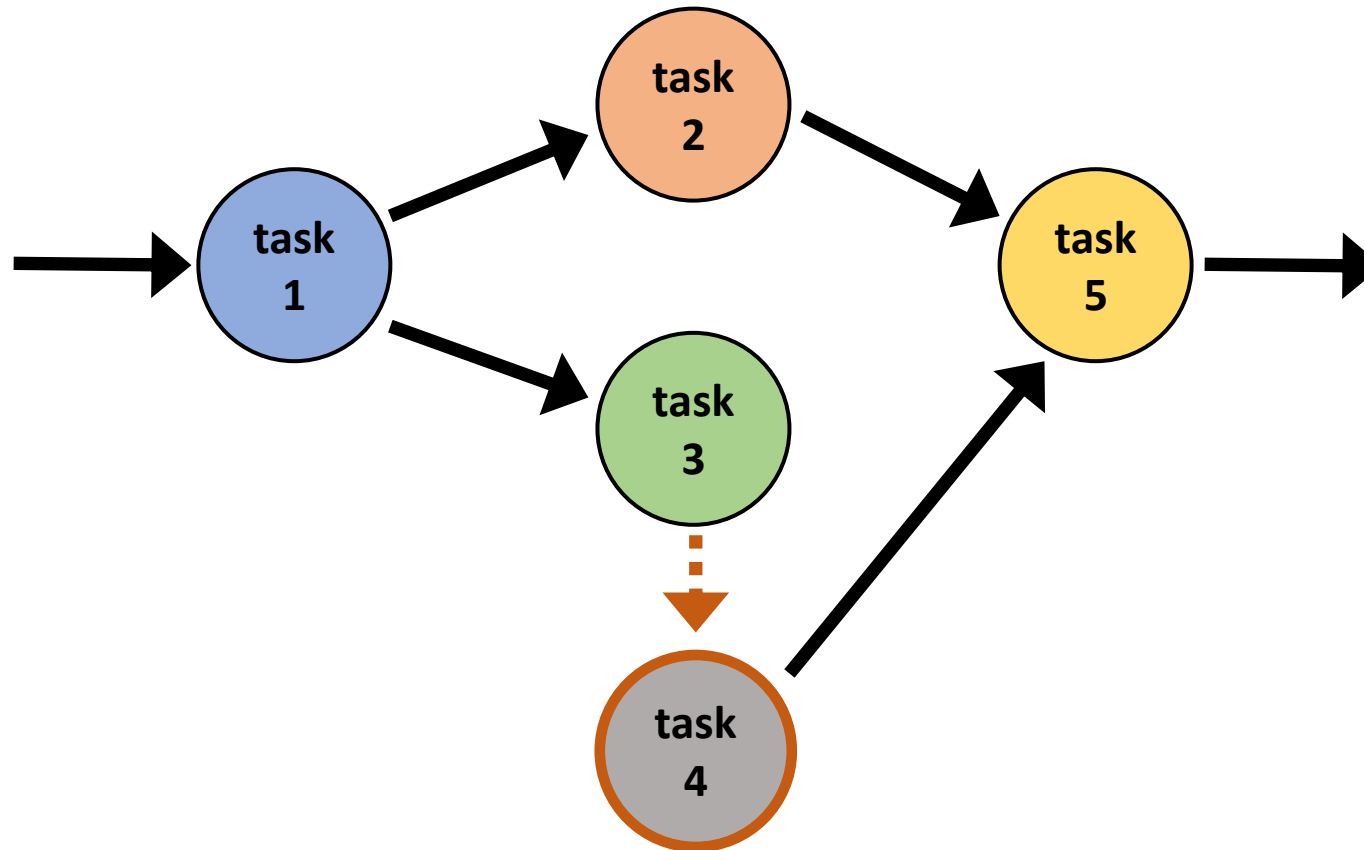
- Speculative task parallelism



Parallelism and Concurrency

e.g., Parallel but not concurrent:

- Speculative task parallelism



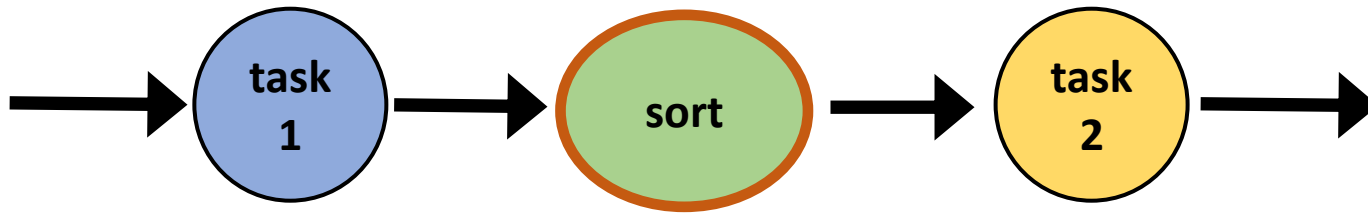
Parallelism and Concurrency

e.g., Parallel but not concurrent:

Parallelism and Concurrency

e.g., Parallel but not concurrent:

- Hardware surrogate execution [Pyla et al., OOPSLA 2011]



Parallelism and Concurrency

e.g., Parallel but not concurrent:

- Hardware surrogate execution [Pyla et al., OOPSLA 2011]

